

7.3 Trigonometric Substitution - continued

Sometimes the integrals we use are not explicitly in the form of a pure square root. Does this change our process of integration?

Trigonometric change of variables can be applied to definite integrals, where we use inverse functions to compute integration limits for θ .

Example 3

Evaluate $\int_0^{1/2} \frac{4x^2}{(1-x^2)^{\frac{3}{2}}} dx$

What step of the integration process can we disregard when dealing with definite integrals?

Example 4

Evaluate $\int_0^a x^2 \sqrt{a^2 - x^2} \, dx$

Completing the Square

Suppose we wish to write $x - x^2$ in the form $a^2 - (u(x))^2$. How can we do so?

Sometimes there is no clear choice for x . In these cases we must manipulate our integral.

Example 5

Evaluate $\int_0^1 \sqrt{x - x^2} \, dx$

Extra Example

Try this on your own before watching the videos.

Evaluate $\int \frac{1}{\sqrt{x^2 + 2x + 5}} dx$

How do we know when to use trig substitution?

Does the integrand have to have a square root to use trig substitution?

What do students commonly do incorrectly when implementing trigonometric substitution?

Extra Example

Try this on your own before watching the videos.

Evaluate (*using trigonometric substitution only when appropriate*)

1. $\int \frac{x^2}{\sqrt{4-x^2}} dx$

2. $\int \frac{x}{\sqrt{3x^2-6}} dx$

3. $\int \frac{1}{x^2+3} dx$

Trigonometric substitution can be helpful in evaluating integrals that contain terms of the form $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$ or $\sqrt{x^2 - a^2}$.

$\sqrt{a^2 - x^2}$	$\sqrt{a^2 + x^2}$	$\sqrt{x^2 - a^2}$
Use $x =$	Use $x =$	Use $x =$
$\sin \theta =$	$\tan \theta =$	$\sec \theta =$
$x =$	$x =$	$x =$
$dx =$	$dx =$	$dx =$
$\frac{\sqrt{a^2 - x^2}}{a} =$	$\frac{\sqrt{a^2 + x^2}}{a} =$	$\frac{\sqrt{x^2 - a^2}}{a} =$
$\sqrt{a^2 - x^2} =$	$\sqrt{a^2 + x^2} =$	$\sqrt{x^2 - a^2} =$
$\theta =$	$\theta =$	$\theta =$
$\leq \theta \leq$	$< \theta <$	$\leq \theta <$ if $\left(\frac{x}{a}\right) \geq 1$ $\leq \theta <$ if $\left(\frac{x}{a}\right) \leq 1$